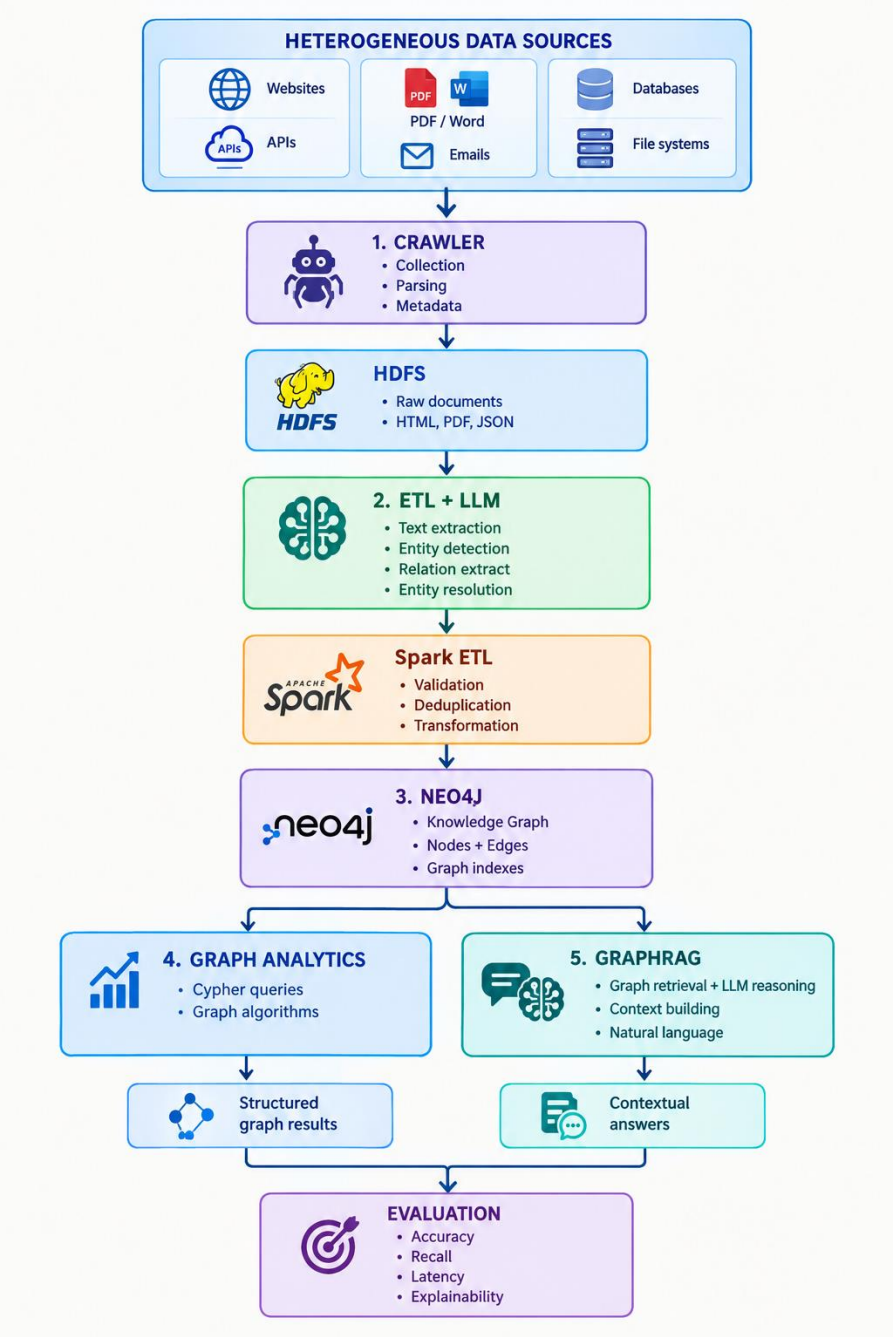
AI-powered knowledge graph analytics platform

The project architecture describes an **AI-powered knowledge graph analytics platform** that combines Big Data, LLM-based information extraction, Neo4j, and GraphRAG. The objective is to collect information from heterogeneous sources, extract entities and relationships, build a knowledge graph, and compare traditional graph analytics with LLM-based graph retrieval and reasoning.

1. Proposed architecture



The key design principle is to maintain **one knowledge graph that supports two different analytical approaches**:

- **Graph Analytics:** direct queries and graph algorithms using Cypher and Neo4j Graph Data Science.
- **GraphRAG:** retrieval of graph information combined with an LLM to answer complex questions in natural language.

This makes the comparison meaningful because both approaches can use the same underlying data.

2. Implementation roadmap

Recommended phases.

Phase	Scope	Deliverable
1	Sample documents, crawler, HDFS	Working ingestion pipeline
2	LLM extraction and entity resolution	Structured entities and relationships
3	Spark transformations and Neo4j ingestion	Populated knowledge graph
4	Cypher queries and graph algorithms	Graph analytics module
5	GraphRAG retrieval and generation	Natural-language graph assistant
6	Ground-truth dataset and comparative tests	Benchmark report
7	API, interface, monitoring, and access control	Integrated application

3. Final recommendation

The strongest contribution would be **a comparative evaluation of structured graph analytics and GraphRAG over a knowledge graph automatically constructed from heterogeneous data sources**.

The main targeted questions could be:

1. How accurately can an LLM extract entities and relationships from heterogeneous documents?
2. How does entity resolution affect the quality of the resulting knowledge graph?
3. How does GraphRAG perform compared with direct Cypher queries for multi-hop and semantic questions?
4. What trade-offs exist between answer quality, completeness, latency, and inference cost?
5. How can source provenance improve the reliability and explainability of generated answers?

One additional consideration is important if the platform analyzes networks of identifiable people: restrict collection to lawful, authorized sources, apply access controls and data-retention policies, and avoid presenting inferred associations as verified facts.

The next practical step is to build a Docker Compose proof of concept containing HDFS, Spark, Neo4j, a Python LLM extraction service, and a GraphRAG service. Start with a small collection of documents and a shared graph schema; then add distributed processing and benchmarking once the end-to-end workflow works.